MT6102 MARITIME ECONOMICS · GROUP INVESTMENT PROPOSAL

USD 250 Million Shipping Investment Proposal

Skip the peak, buy value, lock income: five secondhand ships, three fixed charters, two spot ships and a cash reserve

Group X · Semester 1 2026/27 · Group members and matriculation numbers to be added here (10 members)

Every purchase price, charter rate and freight rate in this deck is the last transacted sale or fixture on the Clarksons Shipping Intelligence Network (SIN). Price series extracted 12 Sep 2026 and re-checked 15 Oct 2026; supply data extracted 10 Oct 2026.

AGENDA

How the proposal is built

1. **The decision**

   How the USD 250m is allocated, why each line is sized that way, and which data supports it. Five ships, four trades, one cash line.

2. **Pricing basis**

   Every purchase price, charter rate and freight rate is a last transacted sale or fixture on Clarksons SIN.

3. **Demand and supply**

   Three demand engines, deliveries against demolition, the 2027 to 2028 delivery wave, and a replacement test against the ageing fleet.

4. **Market data and news**

   Peak signals in large crude tankers, recovery in dry bulk, what changed since 12 September, and our freight-rate call.

5. **Earnings**

   One formula, the base case by vessel, three scenarios with an expected value, and sensitivity.

6. **Shocks and mitigation**

   Five shocks, charter expiry timed against the delivery wave, and the triggers we watch.

7. **Conclusion**

   The same three questions in the same order: how we allocate, why, and what data says so.

Where the USD 250m goes, why, and the data behind it

Allocation	USD m (share)	Why this much	Data behind the decision
2× MR product tankers **5-year-old Eco, 50k dwt. One on a 1-year TC at $30,000/day, one spot**	**100 (40%)**	Closest to replacement cost of the five segments: 5-year-old $50m against newbuild $52m in September (0.96), $52m against $52.5m in October (0.99). Products trade +6.3% in 2027F. Spot earnings $53,741/day in the week of 9 Oct sit above the 1-year TC, so one ship stays open.	Series 47130 / 71267 (prices), 98802 / 534403 (products trade), 543759 (MR Eco earnings). Fixtures PS Imabari to SUNCOR $30,000 and Challenge Pollux to ExxonMobil $27,000, 17 Aug 2026.
1× Aframax crude tanker **2018 Japan-built, 110k dwt. 3-year TC at $40,000/day**	**75 (30%)**	Lowest orderbook of any segment: 8.4% of dwt, 62 ships on order against 223 over 20 years old. A 3-year fixture at $40,000 exists (Torm Genesis to AET, 31 Aug), so income is certain through the 2027 to 2028 delivery wave. No Suezmax: 5-year-old $130m against newbuild $90m in October, 1-year TC $140,250 against 3-year $64,375.	Series 534456 (orderbook), 531168 (earnings), 87302 / 53070 and 48989 / 49108 (Suezmax signals), datasets L13 to L16. Sales Southern Reverence and Pusaka Borneo at $75.0m.
2× Kamsarmax bulkers **82k dwt, Japan-built 2019 and 2015. One on a 12-month TC at $22,000/day, one spot for the Q4 grain season**	**70 (28%)**	Recovery, not peak: BDI 3,521 on 10 Sep and 2,801 on 13 Oct; 5-year-old price 9% above newbuild, against 23% to 44% for Suezmax. Orderbook 13.5%; 2025 deliveries 143 against demolition 14; 2027 deliveries 171 against 618 ships over 20 years old. Guinea bauxite +29% y/y, Simandou iron ore ramping up.	Series 17259 (BDI), 534450, 540721, 557125, 534400 / 547349, 533288. Sales Aquavita Aim $38.0m and Medi Positano $32.0m. Fixtures Intensity to CNR $22,000, XH Hope to Cargill $23,000, coal DBCT to Vizag $25.60/t (SAIL).
Cash reserve **Counter-cyclical fund and working capital**	**5 (2%)**	Dry powder for when tanker secondhand prices fall back below newbuild. Equals five months of fleet OPEX ($36,560/day). Fixed charters add about $2.7m a month.	OPEX series 540687 / 540689 / 540683; scrap price series 542047.
Total	**250 (100%)**	Tankers 70%, dry bulk 28%, cash 2%. 60% of ship-days fixed, 65% of base income. Earnings before OPEX: Bear 17.9%, Base 19.9%, Bull 23.7%.	MT6102_Earnings_Model.xlsx

Source: Clarksons SIN series and fixture register, extracted 12 Sep 2026, spot anchors week of 9 Oct 2026, supply data 10 Oct 2026. Full series list in the appendix.

PART 1 · THE DECISION

Capital allocation: five ships and one cash line, exactly USD 250m

Asset	Specification	Price USD m	Basis: last transacted sale or assessment	Employment
MR #1	50k dwt, 5-year-old Eco	50	SIN 5-year-old MR assessment, series 47130	1-year TC at $30,000/day
MR #2	50k dwt, 5-year-old Eco	50	Same basis	Spot, clean products
Aframax	110k dwt, 2018 Japan-built	75	Southern Reverence and Pusaka Borneo, both $75.0m (May and Apr 2026)	3-year TC at $40,000/day
Kamsarmax #1	82k dwt, 2019 Oshima (Japan)	38	Aquavita Aim, $38.0m (10 Aug 2026)	12-month TC at $22,000/day
Kamsarmax #2	82k dwt, 2015 Tsuneishi (Japan)	32	Medi Positano, $32.0m (10 Aug 2026)	Spot, Q4 grain season
Vessels		**245**		
Cash reserve	**Counter-cyclical fund**	**5**		**Buy the collapse**
Total		**250**		

Financing in one line

The USD 250m is treated as equity, so there is no interest and no repayment schedule to meet in a weak year. Banks lend 60 to 80 percent against 5-year-old tonnage if the board later wants leverage; the Aframax with its 3-year charter is the natural first asset to mortgage or to sell and lease back.

All five ships are Japan or Korea built

They sit outside the US Section 301 port-fee regime whose suspension expires around 10 November 2026. The market already prices this: a 2015 Japan-built Kamsarmax sold for $32.0m against $29.25m for a China-built sister (Presinge), a $2.75m premium for the same insurance.

Source: SIN S&P Register and 5-year-old valuation series 47130, extracted 12 Sep 2026 (October assessment $52m). Screenshots of every record are on file.

Buy secondhand, not charter-in, not newbuild

Charter-in rejected

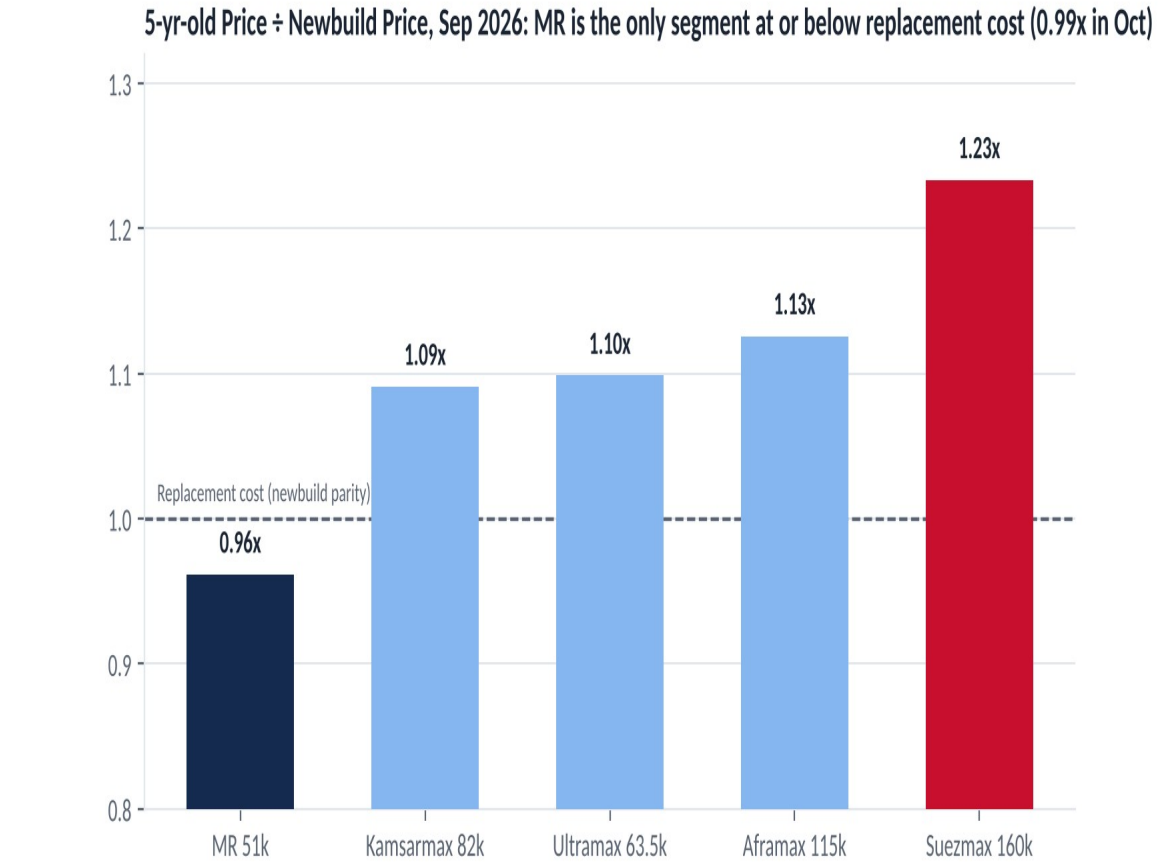
Period rates sit at cycle highs, so chartering in locks a cost at the top. A chartered-in MR costs $30,000/day and earns $27,000 in the base case and $22,950 in the bear case. Only the bull case pays, and there is no asset to resell.

Newbuild rejected

Yards are full to 2029, so the earliest delivery lands inside the 2027 to 2028 delivery wave. Newbuild prices are at decade highs and a newbuild earns nothing during the 12-month window.

5-year-old Eco chosen over 10-year-old

Cost per remaining service year is lower (MR $2.50m against $2.67m a year), the ships are Eco specification for EU ETS and FuelEU, and 5-year-old prices sit at or just above newbuild in our segments, while Suezmax trades at 1.44 times newbuild.



Source: Clarksons Research, Shipping Intelligence Network [A1-A6 series: 47130/542021/540721/38981/87302 vs 71267/23357/70747/35905/53070]. Extracted 12 Sep 2026.

Source: SIN series 47130 / 542021 / 540721 / 38981 / 87302 (5-year-old prices) and 71267 / 23357 / 70747 / 35905 / 53070 (newbuild prices), September 2026; 10-year-old prices dataset L3.

Which trades our ships go into

2× MR product tankers

Clean products: Middle East and India refineries to East Africa and South East Asia

Refining capacity is moving east, so each tonne travels further. Products trade +6.3% in 2027F and tonne-miles +6.5%. One ship fixed to SUNCOR for a year, one trading spot where earnings are $53,741/day.

1× Aframax crude tanker

Crude oil worldwide, routed by an oil-major-class charterer under a 3-year charter

The lowest orderbook of any segment (8.4% of dwt). We sell certainty rather than spot exposure: $40,000/day for three years, through the delivery wave and past any ceasefire.

1× Kamsarmax (fixed)

Bauxite Guinea to China and Pacific coal, on a 12-month charter to CNR

Guinea exports +29% y/y; the Guinea to China run is the longest haul in dry bulk, so volume growth counts twice. Kamsarmax is the largest ship Port Kamsar can take.

1× Kamsarmax (spot)

Grain US Gulf and Black Sea to Asia in the Q4 season, then bauxite and coal

Harvest seasonality lifts US Gulf exports from September; 2026 is a bumper grain year. If the Black Sea stays closed the ship shifts to bauxite and coal, where two recent coal fixtures confirm the cargo.

Source: SIN series 98802 / 534403 (products trade), 534456 (crude Aframax orderbook), 557125 (Guinea bauxite), 534400 / 547349 (grain); SIN news #224049, #227870, #191021.

Pricing basis I: secondhand prices from last transacted sales

Date	Vessel	Type, dwt, built	Price	Used for
01 May 2026	Southern Reverence	Aframax, 108,534 dwt, 2018 Tsuneishi (Japan)	**$75.0m**	Our Aframax at $75m
24 Apr 2026	Pusaka Borneo	Aframax, 108,667 dwt, 2018 Tsuneishi (Japan)	**$75.0m**	Second comparable at the same price
10 Aug 2026	Aquavita Aim	Kamsarmax, 82,192 dwt, 2019 Oshima (Japan)	**$38.0m**	Our Kamsarmax #1
10 Aug 2026	Medi Positano	Kamsarmax, 81,661 dwt, 2015 Tsuneishi (Japan)	**$32.0m**	Our Kamsarmax #2
11 Aug 2026	Presinge	Kamsarmax, 81,886 dwt, 2015 Tsuneishi Zhoushan (China)	**$29.25m**	Control: the $2.75m discount for China-built tonnage prices the port-fee risk
Sep 2026	MR 5-year-old benchmark	SIN assessment, series 47130 (October: $52m)	**$50.0m**	Our two MRs

Two same-type, same-age Aframax sales at an identical $75.0m make that price firm. The two Kamsarmaxes are benchmarked to two Japan-built sales on the same day. The Presinge control shows what the market charges for the port-fee insurance we are buying.

Source: SIN S&P Register (Sales by vessel type), extracted 12 Sep 2026; all 18 records in the appendix with screenshots on file.

Pricing basis II: time-charter rates from last transacted fixtures

Date	Vessel	Charterer	Period and rate	Used for
17 Aug 2026	PS Imabari (MR, 52,838 dwt, 2022)	SUNCOR	**1 year at $30,000/day**	MR #1 fixed rate
17 Aug 2026	Challenge Pollux (MR, 50,068 dwt, 2017)	EXXONMOBIL	**1 year at $27,000/day**	MR #2 base-case spot proxy
31 Aug 2026	**Torm Genesis (Aframax, 119,456 dwt)**	**AET**	**3 years at $40,000/day**	**Our Aframax 3-year charter**
07 Sep 2026	Intensity (Kamsarmax, 82,000 dwt, 2026)	CNR	**12 months at $22,000/day**	Kamsarmax #1 fixed rate
03 Sep 2026	XH Hope (Kamsarmax, 84,998 dwt, 2022)	CARGILL	**10 to 12 months at $23,000/day**	Kamsarmax #2 base-case spot proxy

Why these charterers

Oil majors, a Canadian integrated producer and the largest grain house. Counterparty risk on the fixed 60% of ship-days is as low as the market offers, and a long charter is itself bankable security.

What the October refresh adds

The SIN 1-year MR Eco assessment rose from $29,000 in September to $43,000 in October, and MR spot earnings reached $53,741/day. The $30,000 fixture is kept as the last transacted rate; the final data cut will replace it with the October fixture, which only raises the fixed leg.

Source: SIN Fixtures Register (Timecharter), extracted 12 Sep 2026; SIN 1-year MR Eco TC assessment series 541552, October 2026 value $43,000/day.

Pricing basis III: spot freight rates from last transacted voyage fixtures

Date	Route and cargo	Rate	Charterer	Role in the proposal
10 Sep 2026	Coal 80,000 t, DBCT (Australia) to Vizag (India)	**$25.60/t**	SAIL	Kamsarmax spot leg; converts to about $27,400/day TCE at September bunker prices, above the $23,000 base assumption
09 Sep 2026	Coal, Newcastle (Australia) to Hadong (Korea)	**$23.21/t**	KEPCO	Second coal fixture, same week
10 Sep 2026	Gasoline 37,000 t, Baltic to US Gulf (Dee4 Juniper)	**WS105**	VITOL	MR spot leg, Atlantic
10 Sep 2026	Gasoil 35,000 t, Singapore to Balikpapan (Pacific Cobalt)	**$1.2m lump sum**	SEARIVER	MR intra-Asia evidence
03 Sep 2026	Crude 130,000 t, West Africa to Mediterranean (Sea Amethyst)	**WS210**	SHELL	Context: Suezmax spot at two to three times normal
10 Sep 2026	Crude, Brazil to Chile (Barcelona Spirit)	**WS388**	PETROBRAS	Context: extreme rates we deliberately avoid

Three price families, 17 transactions, all last done: purchase prices, charter rates and freight rates. Our spot exposure sits in coal and clean products, outside the war-premium crude market.

Source: SIN Fixtures Register (Voyage and spot), extracted 12 Sep 2026; voyage TCE calculation in MT6102_Earnings_Model.xlsx, sheet Voyage_TCE.

Demand: three trades, three different engines

+29%

Guinea bauxite exports, Jan to Apr 2026 y/y. Simandou iron ore now shipping; the Guinea to China haul ties up more ships per tonne.

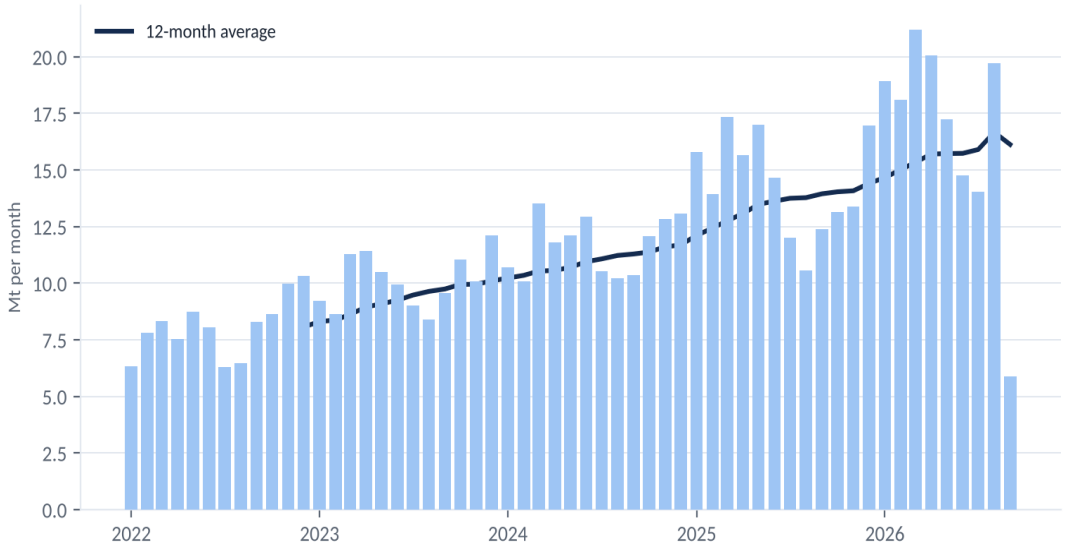
+6.3%

Seaborne products trade, 2027 forecast; tonne-miles +6.5%. Refining capacity shifting east lengthens product hauls.

−7.3%

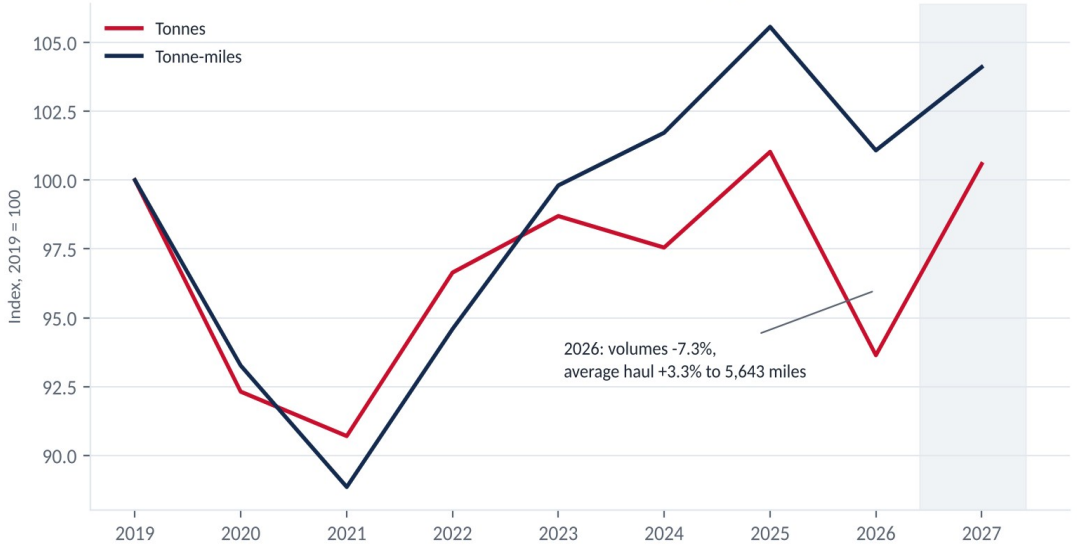
Seaborne crude volumes 2026, yet average haul +3.3%. Hormuz and Red Sea rerouting prop up tonne-miles: lock income with a 3-year charter.

Guinea Bauxite Exports (Mt/month): +29% y/y Jan-Apr 2026, plus Simandou iron ore ramp-up



Source: Clarksons Research, Shipping Intelligence Network [557125]; Clarksons SIN News Article 224049. Extracted 12 Sep 2026. Sep-26 partial month.

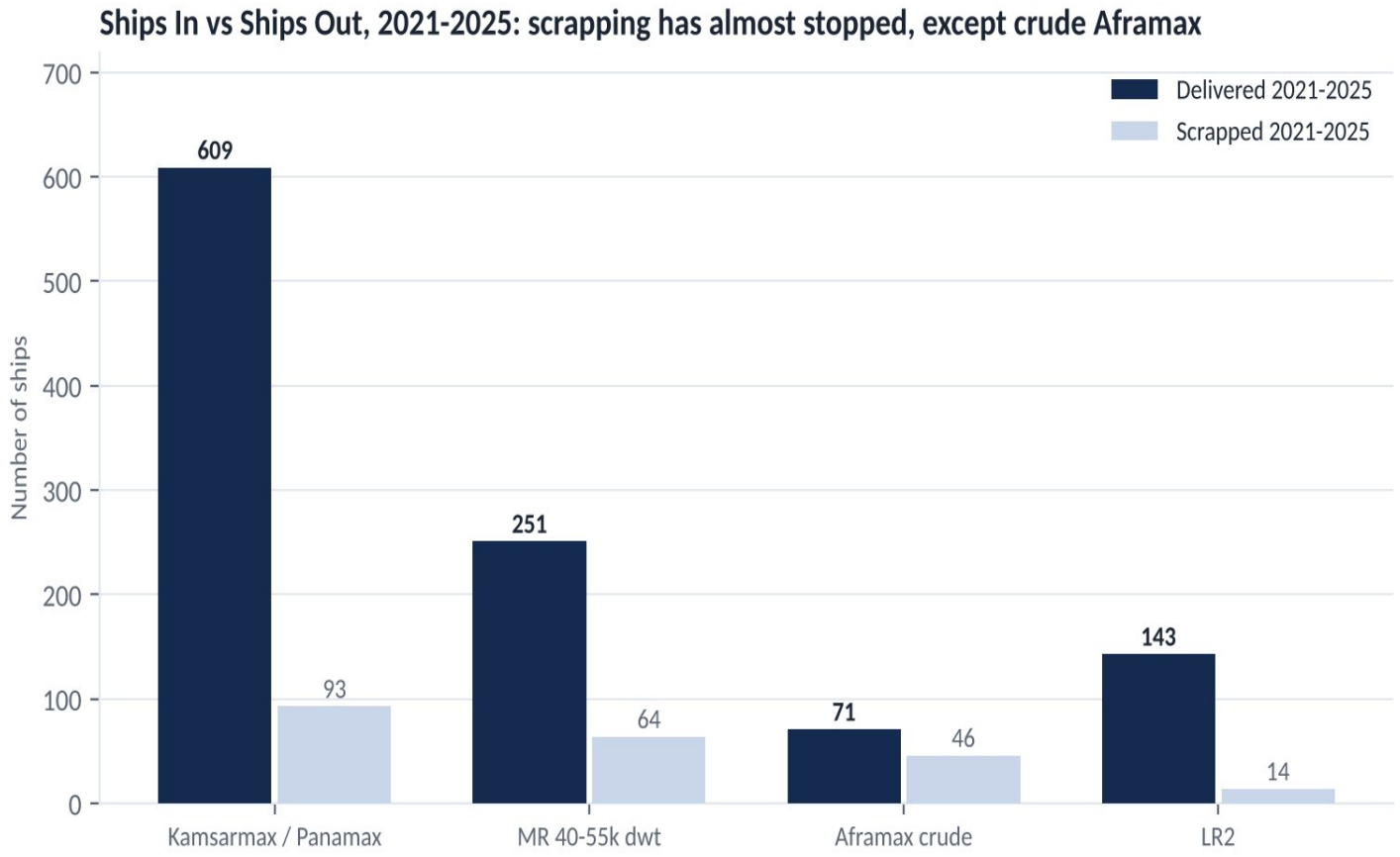
Crude Trade, Indexed (2019 = 100): distance, not volume, is carrying demand



Source: Clarksons Research, Shipping Intelligence Network [98801, 534402]; haul = tonne-miles / tonnes (Week 5 method). Extracted 12 Sep 2026.

Source: SIN series 557125 (Guinea bauxite exports), 98802 / 534403 (products trade), 98801 / 534402 (crude trade and tonne-miles); SIN news #224049, #227870, #191021, #219145. Extracted 12 Sep 2026.

Supply today: ships arrive, almost none leave



Source: Clarksons Research, Shipping Intelligence Network [L13: 30301/540953/29984/39696; L14: 30291/540965/29967/39683]. Crude Aframax = Aframax tanker minus LR2. Extracted 10 Oct 2026.

93 vs 609

Kamsarmax and Panamax scrapped against delivered, 2021 to 2025: only about 15% of new ships were offset.

3 in, 8 out

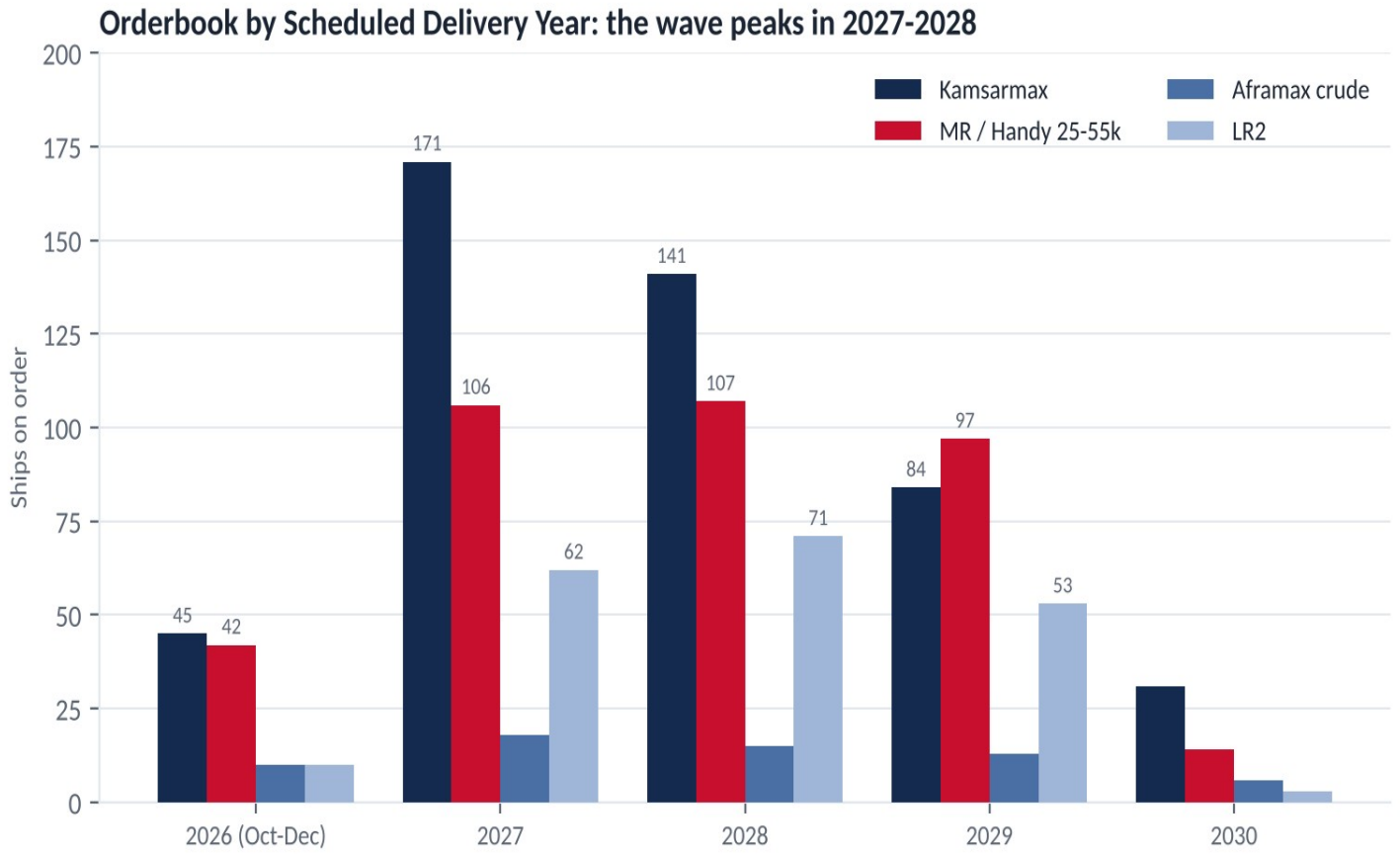
Crude Aframax in 2025: the fleet actually shrank.

153 vs 6

Kamsarmax and Panamax delivered against scrapped, 2026 year to date: scrapping has almost stopped.

Source: SIN deliveries series 30301, 540953, 29984, 39696 and demolition series 30291, 540965, 29967, 39683, extracted 10 Oct 2026. Aframax crude = Aframax tanker series minus LR2. 2026 = year to date.

Supply ahead: a delivery wave peaks in 2027 to 2028



Source: Clarksons Research, Shipping Intelligence Network Orderbook > Delivery Schedule [L15]. By number of ships; MR includes 1 unit due 2031. Extracted 10 Oct 2026.

Orderbook as % of current fleet (ships)

Kamsarmax
472 / 3,487 ships
13.5%

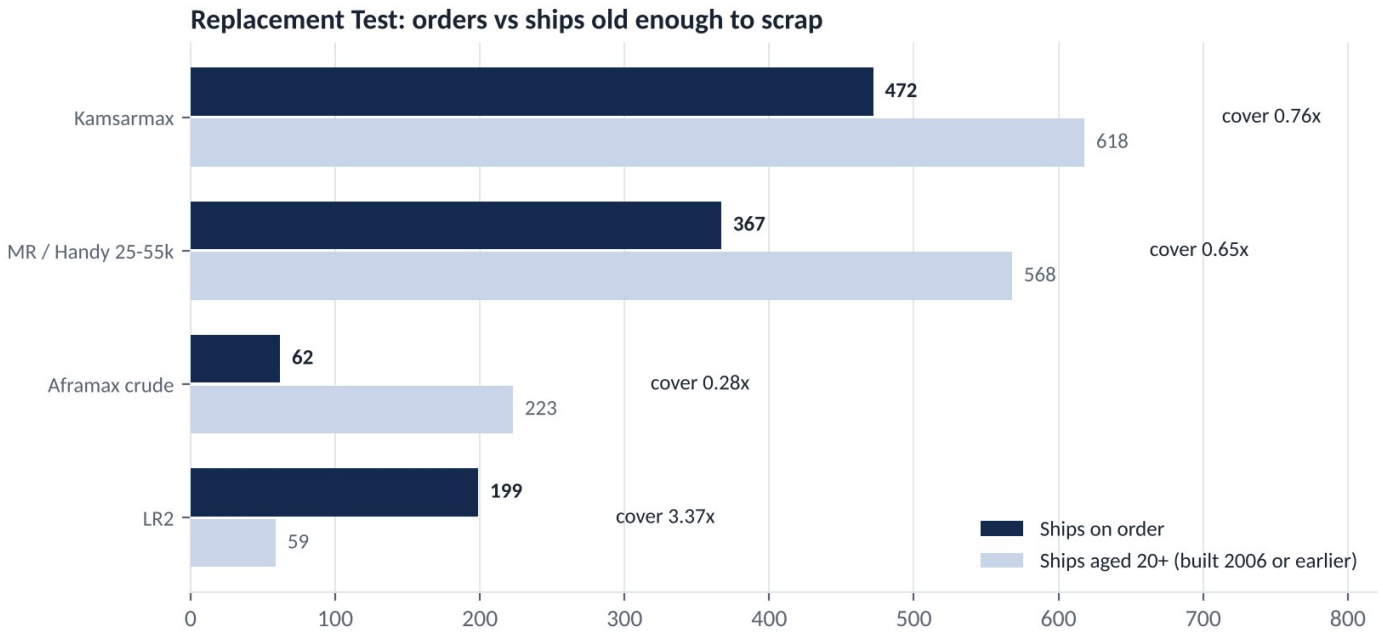
MR
367 / 2,351 ships
15.6%

Aframax crude
62 / 696 ships
8.9%

LR2
199 / 572 ships
34.8%

Source: SIN Orderbook, Delivery Schedule and Fleet Register, extracted 10 Oct 2026. 2026 = deliveries still due Oct to Dec. MR = MR/Handy 25 to 55k dwt (incl. one ship due 2031). By number of ships.

Replacement test: new ships against the ageing fleet



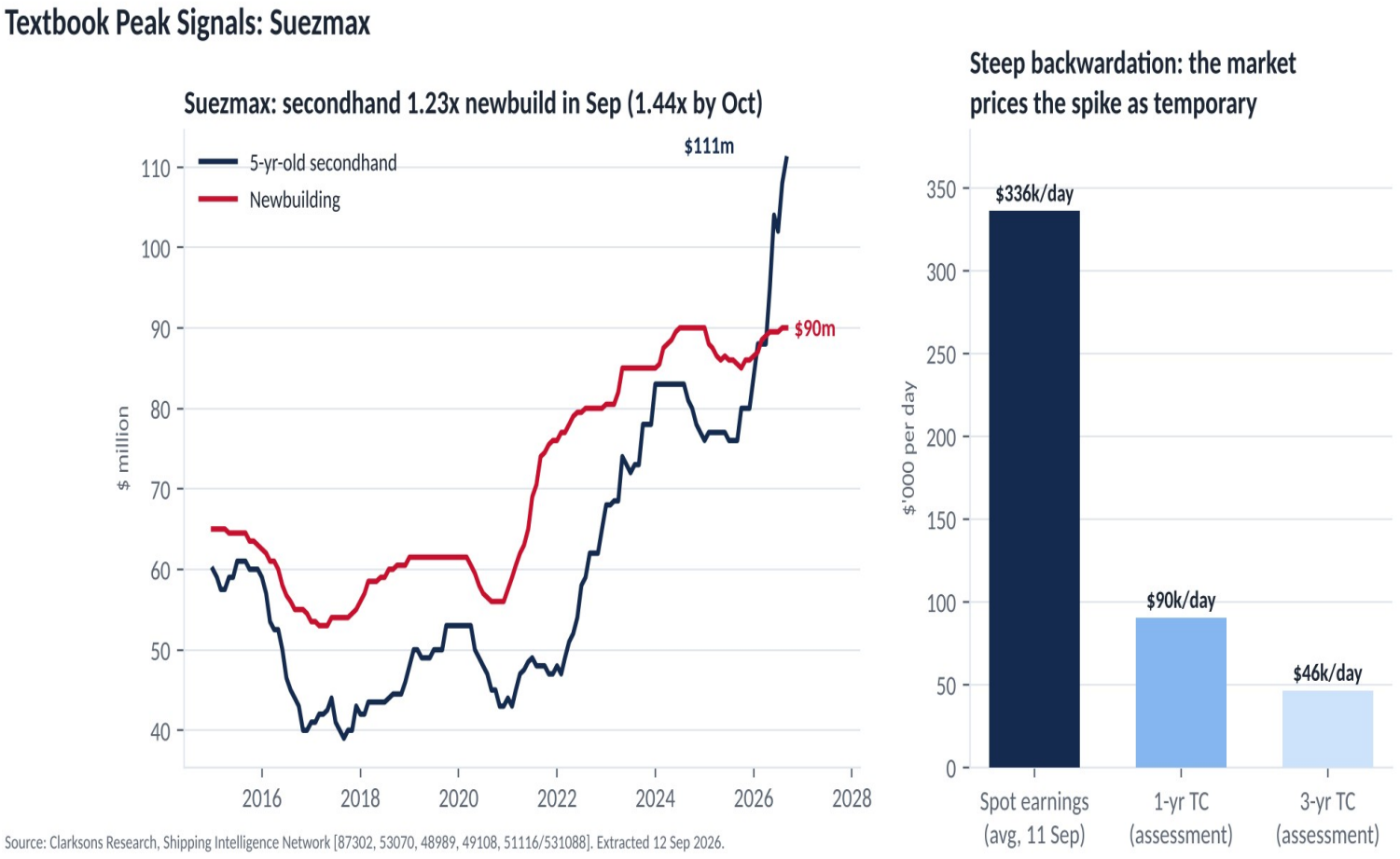
Source: Clarksons Research, Shipping Intelligence Network Orderbook Delivery Schedule [L15] and Fleet Register age profile [L16]. Cover = orderbook / ships aged 20+. Extracted 10 Oct 2026.

Segment	Cover	Verdict and our action
Kamsarmax	**0.76x**	Balanced, 2027 wave: buy 2, fix 1 on a 1-year TC
MR	**0.65x**	Balanced, 2027 to 2028 wave: buy 2, fix 1 on a 1-year TC
Aframax crude	**0.28x**	Tightest supply: buy 1 on a 3-year TC
LR2	**3.37x**	Oversupply, can switch to crude: avoid; the 3-year TC protects our Aframax

Ships live about 25 years, so the 20-year-plus cohort is the pool that scrapping draws from. Kamsarmax, MR and crude Aframax orderbooks are all smaller than that pool; LR2 orders are 3.4 times it. Clarksons' 9 October note on the recycling overhang confirms that the deferred scrapping will come.

Source: SIN Orderbook Delivery Schedule and Fleet Register age profile, extracted 10 Oct 2026. Cover = orderbook divided by ships aged 20 years and over; below 1.0x the orderbook can largely replace scrapping once it resumes.

Skip Suezmax: three peak signals, all louder in October



Source: Clarksons Research, Shipping Intelligence Network [87302, 53070, 48989, 49108, 51116/531088]. Extracted 12 Sep 2026.

1.44x

5-year-old Suezmax against newbuild in October ($130m against $90m); 1.23x in September. Secondhand above replacement cost is the textbook over-trading signal.

$140k vs $64k

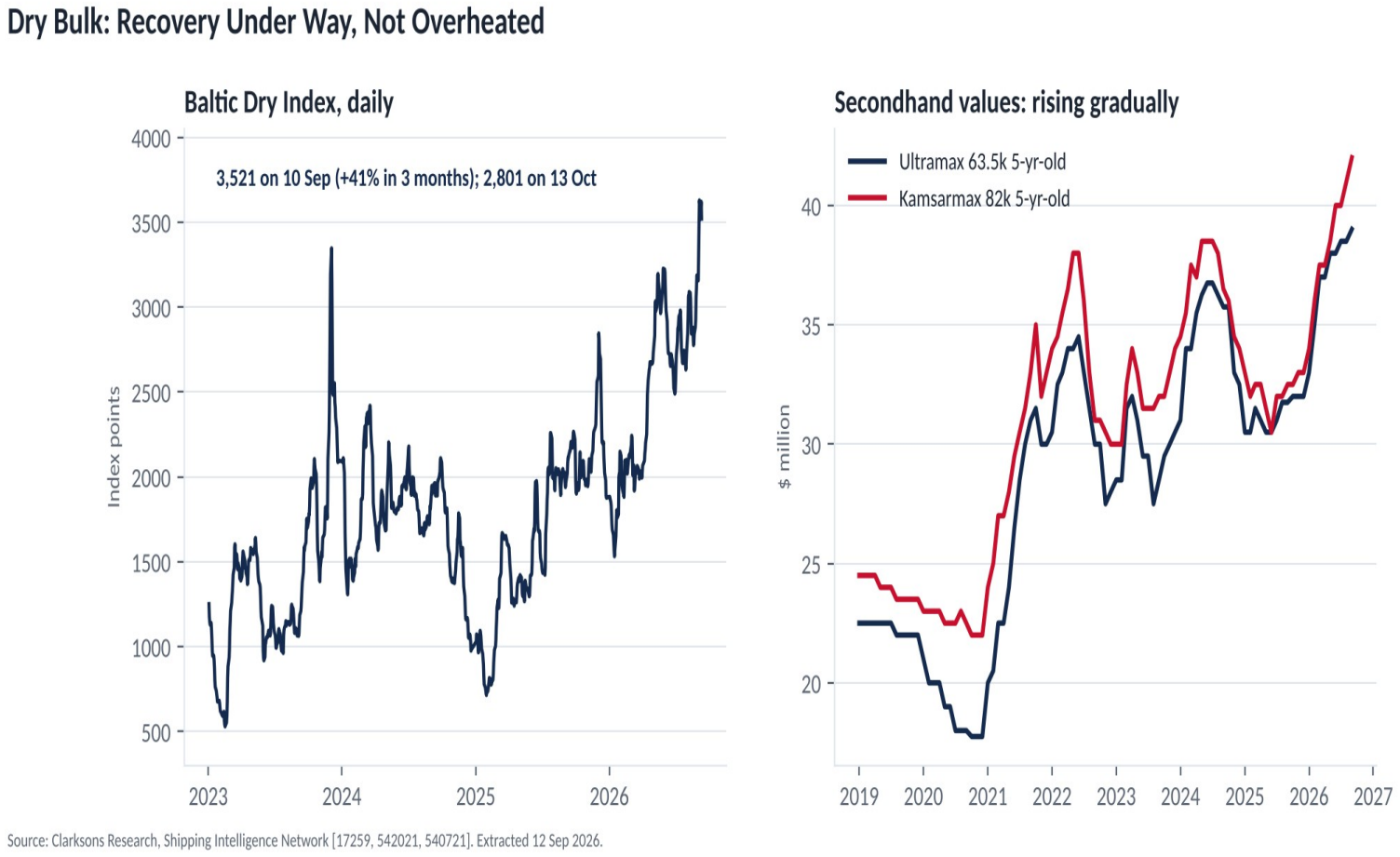
1-year against 3-year TC in October. Spot earnings $718,165/day. The market itself prices today's rates as temporary.

2 to 7 a week

Crude tankers transiting Hormuz (Aug to Oct). Rates reflect a supply shock, and 20% of crude tanker capacity is sanctioned: both reverse quickly.

Source: SIN series 87302 / 53070 (5-year-old and newbuild prices), 48989 / 49108 (1-year and 3-year TC), 531088 (earnings), dataset L6 (Hormuz transits), L11 (sanctioned fleet); news #221824, #227050, #228241. Chart shows September data; October values in the tiles.

Buy value: dry bulk is recovering, MR sits at replacement cost



2,801

BDI on 13 Oct, down 20% from the 3,521 peak of 10 Sep. Strong but off the high: the recovery stage, not the peak.

1.09x

5-year-old Kamsarmax against newbuild ($42m against $38.5m). A 9% premium, against 44% for Suezmax.

0.99x

5-year-old MR against newbuild in October (0.96x in September). The lowest ratio of the five segments: we buy at replacement cost.

Source: SIN series 17259 (BDI, daily to 13 Oct 2026), 542021 / 540721 (5-year-old bulker prices), 47130 / 71267 (MR prices); news #225971, #228048, #219145, #191021.

What changed since 12 September, and our freight-rate call

Indicator	12 Sep	October	Read
Suezmax 5-year-old price, $m	111	130	Peak signal stronger
Suezmax 1-year TC, $/day	90,250	140,250	Backwardation steeper
MR 1-year TC assessment, $/day	29,000	43,000	Fixed leg has upside
MR Eco spot earnings, $/day	35,804	53,741	Bull case raised
Kamsarmax Eco spot, $/day	24,484	23,389	Stable, above the $22,000 fix
BDI	3,521	2,801	Off the high, trigger 2,000 not hit
Singapore VLSFO, $/t	811.69	874.94	Charterers pay fuel on our fixed ships

- Torm ordered six MR2s on 26 Aug: new MR supply, but delivery after 2028.
- VLCC spot about $1.3m/day in early October: the war premium is still intact.
- The US and China port-fee suspension expires around 10 November 2026, inside our presentation window.
- Clarksons, 9 Oct: recycling has run below 0.5% of the fleet a year for five years; an overhang of old ships is building.

Our rate call for the next 6 to 12 months

MR

Firm into the first half of 2027 (spot $53,741 against our $30,000 fix), softer once 2027 deliveries land. So one ship fixed for a year, one kept open.

Aframax

Today's rate is a war premium that falls on a ceasefire or sanctions relief. So the whole ship is fixed for three years at $40,000.

Kamsarmax

Firm through the Q4 grain season, then pressured by 171 deliveries in 2027. The spot ship is reviewed against the BDI trigger in Q1 2027.

Source: SIN series re-pulled 15 Oct 2026 (October monthly assessments, weekly earnings to 9 Oct, BDI to 13 Oct); SIN news #229922 (9 Oct), #217187, #228241, #224049, #227789; USTR notice on the port-fee suspension.

Method: one formula, three assumptions

Annual earnings = daily rate × 350 operating days × number of ships

Return on capital = fleet earnings ÷ USD 250m. Forecast window October 2026 to September 2027.

350 operating days

365 days less 15 days of off-hire for repairs, surveys and the first special survey due on the 5-year-old MRs. Each month of delivery delay removes one twelfth of that ship's year.

Before OPEX, at TCE level

The brief asks for revenue, not operating costs. On a time charter the charterer pays bunkers, port and canal costs; on the spot ships the TCE already nets them out. OPEX is kept as a reference line for questions.

Commission 0%

Set to zero so the slide figures match gross hire. At the industry norm of 3.75% (address plus brokerage) the base case is $47.8m, or 19.1%, with every scenario reduced by the same share.

Source: MT6102_Earnings_Model.xlsx (sheets Notes and Inputs); OPEX reference series 540687 / 540689 / 540683.

Base case by vessel: USD 49.7m, 19.9% on capital

Vessel	Employment	Rate, $/day	Annual earnings, $m	Share of fleet income
MR #1	1-year TC, SUNCOR	30,000	10.5	21%
MR #2	Spot, base proxy Challenge Pollux fixture	27,000	9.45	19%
Aframax	3-year TC, AET benchmark	40,000	14.0	28%
Kamsarmax #1	12-month TC, CNR	22,000	7.7	15%
Kamsarmax #2	Spot, base proxy XH Hope fixture	23,000	8.05	16%
Fleet	**60% of ship-days fixed**	**142,000**	**49.7**	**100%**

Each line is the daily rate times 350 days. MR #1: 30,000 × 350 = $10.5m. Aframax: 40,000 × 350 = $14.0m. The two spot ships use the latest one-year fixtures as a conservative proxy; the coal voyage fixture converts to about $27,400/day, above the $23,000 used here.

$142,000

fleet income per day in the base case

19.9%

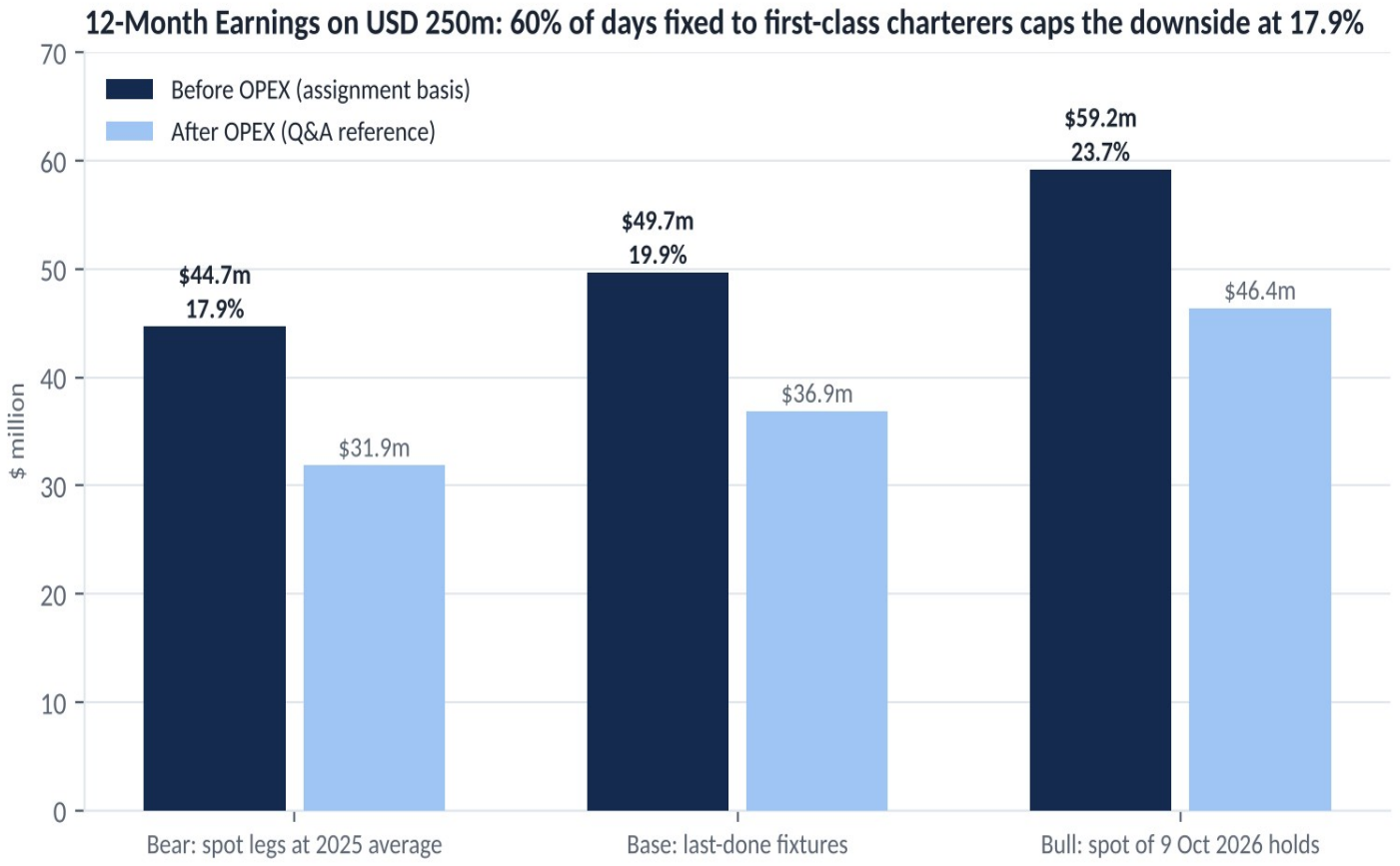
return on USD 250m before OPEX; 14.8% after OPEX of $12.8m

65%

of base income comes from the three fixed ships ($32.2m), identical in every scenario

Source: MT6102_Earnings_Model.xlsx, sheets Inputs and Forecast; rates from the fixtures on the Part 2 slides.

Three scenarios: the floor is 17.9% with 60% of days fixed



Basis: last-done fixtures (SIN Fixtures Register, Aug-Sep 2026); Bull = SIN earnings series week of 9 Oct 2026; OPEX = Clarksons series [540687/540689/540683]; 350 operating days/ship.

Scenario	Spot legs, $/day	Earnings	Return
Bear	22,950 / 12,880	$44.7m	17.9%
Base	27,000 / 23,000	$49.7m	19.9%
Bull	53,741 / 23,389	$59.2m	23.7%
Expected (30 / 50 / 20)		**$50.1m**	**20.0%**
Base after OPEX		**$36.9m**	**14.8%**

Why the bear case still works

Only the two spot ships move. Even at 2025 average earnings, the year before the tanker boom, the fleet returns 17.9%. The bull case has already been overtaken by the October market, so our upside is stated conservatively.

Source: Bear = 2025 average of SIN earnings series 543759 (MR Eco) and 533288 (Kamsarmax Eco); Base = last-done 1-year fixtures; Bull = the same series in the week of 9 Oct 2026. Weights are the group's judgement.

Sensitivity: the fleet stays above 16.8% even if both spot ships collapse

MR #2 spot, $/day → Kamsarmax #2 spot, $/day	13,000	18,000	23,000	28,000
15,000	16.8%	17.5%	18.2%	18.9%
20,000	17.5%	18.2%	18.9%	19.6%
25,000	18.2%	18.9%	19.6%	**20.3%**
30,000	18.9%	19.6%	**20.3%**	**21.0%**
35,000	19.6%	**20.3%**	**21.0%**	**21.7%**

Voyage cross-check: coal DBCT to Vizag	
Cargo and rate	80,000 t at $25.60/t, SAIL, 10 Sep 2026
Gross freight	$2.05m
Round voyage	39 days, 4,600 nm each way
Bunkers	848 t VLSFO at $874.94/t, $0.74m
Port costs and commission	$0.16m and 3.75%
Voyage TCE	**about $27,400/day**
Base assumption	$23,000/day: base case is conservative

Reading the grid

Every $1,000/day on both spot ships moves annual earnings by $0.7m (2 ships × 350 days). The worst cell, both ships at $15,000 and $13,000, still returns 16.8%. The return falls below 15% only if the two ships together earn less than $15,100/day.

Source: MT6102_Earnings_Model.xlsx, sheets Sensitivity and Voyage_TCE; Singapore VLSFO September 2026 average $874.94/t (SIN series 542456).

Five shocks, five prepared answers

Shock	Probability signal	Impact on the fleet	Our answer
Middle East ceasefire or sanctions relief, tanker rates collapse	The forward curve already prices it: Suezmax 1-year TC $140k against 3-year $64k	We own no Suezmax; the Aframax 3-year charter is untouched; the MR spot leg falls, and the bear case already uses 2025 averages	Cash plus about $2.7m a month of fixed hire buys tankers once secondhand prices fall below newbuild
US and China port-fee suspension expires around 10 Nov 2026	Dated event; USTR notice, news #217187 and #227049	China-built ships calling at US ports pay fees	All five ships are Japan or Korea built; the premium is already in our purchase prices
Black Sea escalation blocks grain exports	Attacks on ships and ports continued through August and October, news #227789	Kamsarmax #2 loses one cargo source	Shift to Guinea bauxite and Australian coal; two September coal fixtures prove the cargo is there
Global recession	World seaborne trade already −0.4% in 2026	All three trades cool at once	60% of ship-days fixed, floor 17.9%; both spot ships can be fixed for a year within weeks
Carbon rules tighten (EU ETS, FuelEU, IMO)	EU ETS review opened, news #226856; FuelEU limits step up from 2030	Older, less efficient ships lose value and charterers	Only Eco ships under 10 years old; on time charter the charterer carries fuel and allowances

Source: SIN news #217187, #227049 (port fees), #227789 (Black Sea), #226856 (EU ETS review), #228241; USTR notice on the Section 301 suspension; SIN TC assessment series 48989 / 49108.

Charter expiry is timed against the delivery wave



MR #1 and Kamsarmax #1 expire around September 2027, before the wave peaks: renew or go spot one quarter ahead. The Aframax charter runs to 2029, across the whole wave; LR2 orders at 3.37 times the old LR2 fleet are the hidden competitor it hedges.

Source: SIN Orderbook Delivery Schedule (dataset L15), extracted 10 Oct 2026; fixture Torm Genesis 3 years at $40,000/day, 31 Aug 2026.

What we watch, when we act, and two weaknesses we state ourselves

Trigger	Action
Hormuz crude tanker transits above 50 a week for four consecutive weeks	Normalisation signal. Our income is unchanged; start screening secondhand Suezmaxes with a target price below newbuild
BDI below 2,000 for eight consecutive weeks	Fix Kamsarmax #2 on a 12-month charter
MR spot earnings below $20,000/day for eight consecutive weeks	Fix MR #2 on a 1-year charter
Port-fee suspension not renewed in November	No action needed (Japan and Korea built); re-check charterers' US calling plans
Every Friday	One member scans the SIN weekly: fixtures, sanctions lists, ceasefire and tariff news

Weakness 1: MR orderbook

18% of dwt is high. We offset it with the purchase price (at replacement cost), with 2027 products demand (+6.3%) and with one MR fixed for a year.

Weakness 2: a thin cash line

2% of capital is deliberate, because idle cash earns nothing in a strong market. It covers five months of OPEX, and fixed hire adds about $2.7m a month. A variant with one ship fewer and more cash is kept in reserve.

Source: SIN dataset L6 (Hormuz transits), series 17259 (BDI), 543759 (MR Eco earnings), 534440 (product tanker orderbook); OPEX series 540687 / 540689 / 540683.

CONCLUSION

Skip the peak. Buy value. Lock income.

1 **How we allocate**

Tankers 70%, dry bulk 28%, cash 2%: two MRs, one Aframax, two Kamsarmaxes, five ships at or near replacement cost.

2 **Why**

We avoid Suezmax at 1.44 times newbuild and steep backwardation; we buy segments with the smallest orderbooks against their ageing fleets; we fix 60% of ship-days so the worst case still returns 17.9%.

3 **What the data says**

60 time series, 40 last-done transactions and 55 news items from Clarksons SIN, every figure with a series number and a screenshot. Base case 19.9%, expected value 20.0%, floor 17.9% before OPEX.

We do not forecast the market. We lock the structure. Thank you, we welcome your questions.

APPENDIX

Reference pages for questions

A1 Fleet and orderbook detail · A2 Deliveries and demolition by year · A3 Sale and purchase records · A4 Time-charter fixtures · A5 Spot fixtures and voyage conversion · A6 News evidence library · A7 Earnings model structure · A8 Course concepts applied · A9 Glossary · A10 Market backdrop charts

APPENDIX A1

Fleet and orderbook detail, two bases side by side

Segment	Fleet, ships	Fleet, m dwt	Orderbook, ships (Sep)	Orderbook, ships (1 Oct)	Orderbook, m dwt	Orderbook % of fleet (dwt)	Ships aged 20+	Orderbook ÷ 20+ fleet
MR 40 to 55k dwt	1,906	92.5	340	347	16.7	18.0%	568 (MR/Handy 25 to 55k)	0.65x
Aframax incl. LR2	1,257	139.2	254	260	29.0	20.8%	223 crude + 59 LR2	0.28x crude, 3.37x LR2
Suezmax	725	113.8	211	213	33.2	29.1%	not held	not held
Panamax and Kamsarmax	3,467	282.1	463	474	38.6	13.7%	618	0.76x
Handymax and Ultramax	4,574	263.2	568	647	35.8	13.6%	not held	not held

Why two bases

The dwt basis (Sep) is the standard orderbook ratio; the ship basis (10 Oct) is what the replacement test needs. Crude Aframax on its own is 8.4% of dwt and 62 ships; including LR2 the figure is 20.8%, because LR2 newbuilds can trade crude. That is why we hold one Aframax only and fix it for three years.

Source: SIN Fleet Development and Orderbook series (dataset L4, Sep 2026 and 1 Oct 2026), Orderbook Delivery Schedule and Fleet Register age profile (L15, L16, 10 Oct 2026).

APPENDIX A2

Deliveries and demolition by year, ships

Year	Kamsarmax / Panamax delivered	scrapped	MR 40 to 55k delivered	scrapped	Aframax crude delivered	scrapped	LR2 delivered	scrapped
2021	103	9	66	35	25	26	28	8
2022	119	10	51	11	20	7	19	4
2023	130	31	28	4	16	2	22	0
2024	121	21	20	1	7	3	20	0
2025	136	22	86	13	3	8	54	2
2021 to 2025	**609**	**93**	**251**	**64**	**71**	**46**	**143**	**14**
2026 YTD	153	6	84	10	9	5	62	0

Across 2021 to 2025, only 15% of new Kamsarmax tonnage was offset by scrapping and 25% of new MRs; crude Aframax scrapped two thirds of what it took delivery of. In 2026 scrapping has almost stopped in every segment.

Source: SIN deliveries series 30301 / 540953 / 29984 / 39696 and demolition series 30291 / 540965 / 29967 / 39683, extracted 10 Oct 2026. Crude Aframax = Aframax tanker minus LR2. 2026 = year to date.

APPENDIX A3

Sale and purchase records, 18 last-done transactions

Type	Date	Vessel	dwt	Built	Yard (country)	Price $m	Seller	Buyer
MR	14 Aug 2026	On Promise*	50,000	2026	HD Hyundai Medium (Korea)	undisclosed	Onex DMCC	Undisclosed
MR	14 Aug 2026	On Prosper*	50,000	2026	HD Hyundai Medium (Korea)	undisclosed	Onex DMCC	Undisclosed
MR	11 Aug 2026	High Discovery	49,999	2014	HD Hyundai Mipo (Korea)	15.30	Japanese interests	d'Amico clients
MR	07 Aug 2026	Atlantic Eagle	47,128	2007	HD Hyundai Mipo (Korea)	undisclosed	Cido Shipping	Undisclosed
Ultramax	19 Aug 2026	Amaryllis	63,301	2013	Sinopacific Dayang (China)	24.00	Livanos N.G.	Chinese interests
Ultramax	01 Aug 2026	Gramos	61,171	2019	Nantong COSCO KHI (China)	34.50	Helikon	Livanos clients
Ultramax	20 Jul 2026	Seacon Tokyo	66,628	2023	Tsuneishi Zhoushan (China)	41.60	Seacon Shipping	OBE Ships clients
Suezmax	27 Aug 2026	Cape Benat	156,719	2010	Jiangsu Rongsheng (China)	62.50	Pareto Maritime	Lila Global clients
Suezmax	27 Aug 2026	Hedda Knutsen	154,348	2024	COSCO HI Zhoushan (China)	113.00 (TC incl.)	Knutsen NYK	KNOT clients
Suezmax	11 Aug 2026	Bristol	157,077	2024	HD Hyundai Samho (Korea)	undisclosed	Euronav	Undisclosed
Aframax	**01 May 2026**	**Southern Reverence**	**108,534**	**2018**	**Tsuneishi Zosen (Japan)**	**75.00**	**Nissen Kaiun**	**Undisclosed**
Aframax	**24 Apr 2026**	**Pusaka Borneo**	**108,667**	**2018**	**Tsuneishi Zosen (Japan)**	**75.00**	**Pusaka Laut**	**Undisclosed**
Aframax	31 Mar 2026	Penelop	115,091	2006	Daewoo DSME (Korea)	undisclosed	Lundqvist	Undisclosed
Aframax	05 Mar 2026	Green Attitude	112,532	2018	COSCO HI Zhoushan (China)	70.00	Aegean Shipping	Minerva clients
Kamsarmax	14 Aug 2026	BBG Wuzhou	81,895	2016	Tsuneishi Zhoushan (China)	29.00 (auction)	CDB Leasing	Undisclosed
Kamsarmax	**11 Aug 2026**	**Presinge**	**81,886**	**2015**	**Tsuneishi Zhoushan (China)**	**29.25**	**SwissMarine**	**Indian interests**
Kamsarmax	**10 Aug 2026**	**Aquavita Aim**	**82,192**	**2019**	**Oshima (Japan)**	**38.00**	**Olive Shipmanagement**	**Undisclosed**
Kamsarmax	**10 Aug 2026**	**Medi Positano**	**81,661**	**2015**	**Tsuneishi Zosen (Japan)**	**32.00**	**d'Amico Dry**	**Greek interests**

*Source: SIN S&P Register by vessel type, extracted 12 Sep 2026. Prices undisclosed where the register shows none. * = en bloc resale of newbuildings.*

APPENDIX A4

Time-charter fixtures, 16 last-done records

Type	Date	Vessel	dwt	Built	Charterer	Delivery	Period	Rate $/day	Owner
MR	**17 Aug 2026**	**PS Imabari**	**52,838**	**2022**	**SUNCOR**		**1 year**	**30,000**	**Shoei Kisen**
MR	**17 Aug 2026**	**Challenge Pollux**	**50,068**	**2017**	**EXXONMOBIL**		**1 year**	**27,000**	**NYK Line**
MR	31 Aug 2026	SM Falcon	50,017	2017	CLEARLAKE	W Med	2 years	25,500	Korea Line
MR	31 Aug 2026	Stena Imagination	49,689	2016	ESSAR	UK Cont	2 years	27,750	Stena Bulk
Ultramax	26 Aug 2026	W-Lion	63,308	2014	NOVA MARINE	Port Qasim	10 to 13 months	19,000	W Marine
Ultramax	19 Aug 2026	Yuan Hai Qing Han	63,776	2025	CHINALAND	Qinzhou	8 to 10 months	20,750	JIC Leasing
Ultramax	14 Aug 2026	Gramba	61,233	2021	CTM	Dalian	10 to 13 months	19,000	Susesea
Suezmax	17 Aug 2026	Monte Urbasa	156,400	2018	ST SHIPPING	Singapore	1 year	74,500	Ibaizabal
Suezmax	09 Mar 2026	Ithaki Warrior	159,962	2017	TRAFIGURA	Baltic	10 to 14 months	86,000	Polembros
Suezmax	09 Mar 2026	Kapsali	158,546	2011	MERCURIA	Korea	11 to 13 months	80,000	Centrofin
Aframax	24 Aug 2026	TP Spirit	114,139	2016	MERCURIA	Singapore	1 year	53,250	Transpetrol
Aframax	**31 Aug 2026**	**Torm Genesis**	**119,456**	**2011**	**AET**	**Singapore**	**3 years**	**40,000**	**TORM**
Aframax	24 Aug 2026	Hafnia Despina	109,990	2019	MERCURIA	W Med	2 years	50,000	Hafnia
Kamsarmax	**07 Sep 2026**	**Intensity**	**82,000**	**2026**	**CNR**	**Yeosu**	**12 months**	**22,000**	**Aegean Bulk**
Kamsarmax	**03 Sep 2026**	**XH Hope**	**84,998**	**2022**	**CARGILL**	**Qinzhou**	**10 to 12 months**	**23,000**	**Zhejiang Xiehai**
Kamsarmax	27 Aug 2026	Lestari Kuala Lumpur	82,074	2026	CNR	Maoming	9 to 11 months	21,000	Lestari Maritime

Source: SIN Fixtures Register (Timecharter), extracted 12 Sep 2026. Highlighted rows are the five rates used in the earnings model.

APPENDIX A5

Spot fixtures and the voyage-to-TCE conversion

Date	Vessel	Built	Cargo	Load	Discharge	Rate	Charterer
10 Sep 2026	TBN (Kamsarmax)		Coal 80,000 t	DBCT, Australia	Vizag, India	**$25.60/t**	SAIL
09 Sep 2026	TBN (Kamsarmax)		Coal 80,000 t	Newcastle, Australia	Hadong, Korea	**$23.21/t**	KEPCO
03 Sep 2026	Sea Amethyst	2022	Crude 130,000 t	West Africa	Mediterranean	**WS210**	SHELL
10 Sep 2026	Barcelona Spirit	2011	Crude 130,000 t	Brazil	Chile	**WS388**	PETROBRAS
10 Sep 2026	Dee4 Juniper	2022	Gasoline 37,000 t	Ventspils, Baltic	US Gulf	**WS105**	VITOL
10 Sep 2026	Pacific Cobalt	2020	Gasoil 35,000 t	Singapore	Balikpapan, Indonesia	**$1.2m lump sum**	SEARIVER

Step	Value
Gross freight	80,000 t × $25.60 = $2,048,000
Sea days	4,600 nm laden at 12.5 kn + 4,600 nm ballast at 13 kn = 30.1 days; port 9 days; total 39.1 days
Bunkers	30.1 × 27 t + 9 × 4 t = 848 t × $874.94 = $742,000
Port costs and commission	$160,000 + 3.75% × $2,048,000 = $76,800
Net voyage result	$1,069,000
TCE	**$1,069,000 ÷ 39.1 days = about $27,400/day, 19% above the $23,000 base assumption**

Source: SIN Fixtures Register (Voyage and spot), extracted 12 Sep 2026; MT6102_Earnings_Model.xlsx sheet Voyage_TCE; Singapore VLSFO series 542456.

APPENDIX A6

News evidence library: which decision each item supports

Date	Clarksons headline and article number	Point and use	Supports
04 Mar 2026	Middle East Conflict: Update No.3, #221824	Day 5 of the conflict, Hormuz transits collapse: the start of the tanker spike	**skip tanker**
24 Jul 2026	Middle East Update: Disruption Doubling Up?, #227050	Red Sea attacks restart in 2026; VLCC Bab el-Mandeb transits fall; supports longer hauls	**skip tanker**
26 Aug 2026	Sanctioned Vessels Update, #228241	1,004 sanctioned tankers = 20% of crude tanker capacity (Aframax 35%)	**skip tanker, buy Aframax**
28 Jul 2026	Sanctioned Vessels Update, #227115	EU adds 13 ships on 24 Jul: sanctions tighten month by month	**skip tanker**
10 Jul 2026	Middle East Conflict: Shipping Market Update, #226516	How the conflict feeds through to each segment's rates	**skip tanker**
07 May 2026	Guinea Bauxite and Iron Ore Exports Update, #224049	Guinea bauxite exports Jan to Apr +29%; Simandou ramping up	**buy bulk**
25 Jun 2026	Bulkcarrier Markets Enjoying The Summer Sun, #225971	Clarksons confirms bulk at a two-year high: recovery stage	**buy bulk**
21 Aug 2026	Bulkcarrier Market Outlook, August 2026, #228048	Ore plus grain on the demand side, moderate ordering on supply	**buy bulk**
17 Aug 2026	A Bumper Year For Grain Trade (So Far), #227870	2026 a bumper grain year: basis for the Q4 spot Kamsarmax	**buy bulk**
09 Oct 2026	Ship Recycling: Building Up An Overhang, #229922	Recycling under 0.5% of the fleet a year for five years; the overhang backs the replacement test	**buy bulk, buy Aframax**
22 Dec 2025	Seaborne Oil Trade: Bouncing Back, #219145	Products trade recovering since mid-2025: the MR turning point	**buy MR**
28 Jul 2023	Products Trade: In It For The Long Haul?, #191021	Refineries moving east lengthen product hauls: the MR long story	**buy MR**
17 Oct 2025	US And Chinese Port Fees: Taking Stock, #216732	US fees on China-built ships start, China retaliates: build country becomes a cost variable	**ship choice, risk**
30 Oct 2025	US-China Policy: 30th Oct Update, #217187	One-year suspension on both sides; expiry around 10 Nov 2026 inside the presentation window	**ship choice, risk**
24 Jul 2026	US Section 301 Tariff Update, #227049	Latest 301 developments ahead of the expiry	**risk**
21 Jul 2026	EU Review of the Emissions Trading Scheme, #226856	EU ETS review opens: the case for Eco ships under 10 years	**ship choice, risk**
14 Aug 2026	Russia-Ukraine: Vessel and Infrastructure Attacks, #227789	Black Sea attacks hit oil and grain exports: cargo flexibility for Kamsarmax #2	**buy bulk, risk**
13 Oct 2026	BDI 2,801 (SIN series 17259, refreshed 15 Oct)	Dry bulk strong but off the September high	**buy bulk, risk**
26 Aug 2026	Torm orders six MR2s (industry news, not SIN)	A leading owner still invests in MR; delivery after 2028, outside our window	**buy MR, risk**

Source: Clarksons Research news on SIN (article numbers shown), 55 items collected; the 19 below enter the argument. Tags: skip tanker, buy bulk, buy MR, buy Aframax, ship choice, risk.

APPENDIX A7

Earnings model structure: eight sheets, seven steps

Inputs

Every number used, with its source; yellow cells are assumptions

Forecast

Per-ship annual earnings, three scenarios, expected value, return, after-OPEX line

Monthly

The chosen scenario spread over 12 calendar months

Sensitivity

Fleet earnings for any pair of spot rates

Voyage_TCE

Converts the coal fixture into a daily rate

Notes and Sources

Formula, caveats, series list

1. Fix the rates: three ships at their fixture rates, two spot ships at the one-year fixture proxies
2. Scenario values for the two spot ships: bear 2025 average, base fixture, bull latest week
3. Each ship: rate × 350 days
4. Add the five ships: $142,000 × 350 = $49.7m base
5. Divide by $250m: 19.9%; bear 17.9%, bull 23.7%
6. Weight 30 / 50 / 20: expected $50.1m, 20.0%
7. Deduct OPEX of $36,560/day × 350 = $12.8m for the after-OPEX reference: $36.9m, 14.8%

Source: MT6102_Earnings_Model.xlsx; every input cell carries its SIN series number or fixture.

APPENDIX A8

Course concepts applied in the proposal

Concept	Where it is used
Derived demand	Shipping demand follows cargo trade: Part 3 starts from bauxite, products and crude volumes, not from ships
Four shipping markets and the last-done convention	Part 2: every price is the last transacted sale or fixture, the starting point of any negotiation
Cost split under a time charter	Owner pays capital and operating costs, charterer pays voyage costs: the before-OPEX convention and the fuel and allowance exposure of the fixed ships
Cycle stages and peak signals	Secondhand above newbuild, old ships chartered, rates many times OPEX, zero scrapping: the Suezmax diagnosis and the dry bulk recovery call
Ship price determinants, 5 to 6% a year depreciation	The 5-year-old against 10-year-old comparison and the replacement-cost ratio
Demand and supply five-factor model, tonne-miles and average haul	Part 3 structure; crude volumes −7.3% with average haul +3.3%
Supply is inelastic in the short run; build time one to four years	Why a Hormuz closure multiplies rates, and why the 2027 to 2028 wave is already fixed
Scrapping determinants: age, obsolescence, scrap price, earnings, expectations	Why scrapping stopped in 2025 and why the overhang will return
Financiers prefer predictable earnings; debt means fixed repayments	60% of ship-days fixed; all-equity base case; leverage optional later
Regulatory cost incidence and marginal abatement cost	Eco ships abate cheaply; on time charter the charterer carries fuel and allowances

Source: Concepts from the MT6102 lecture series; page references held in the group's working notes.

APPENDIX A9

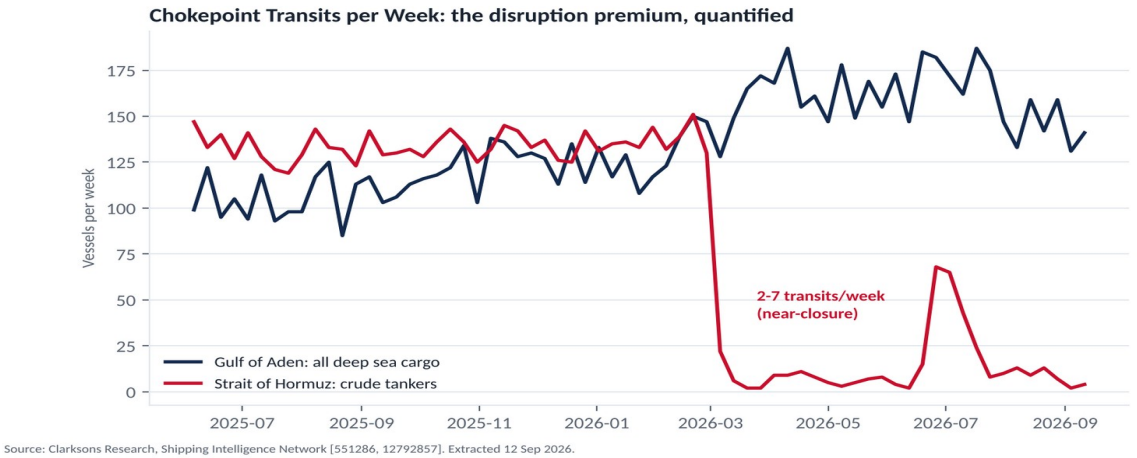
Glossary

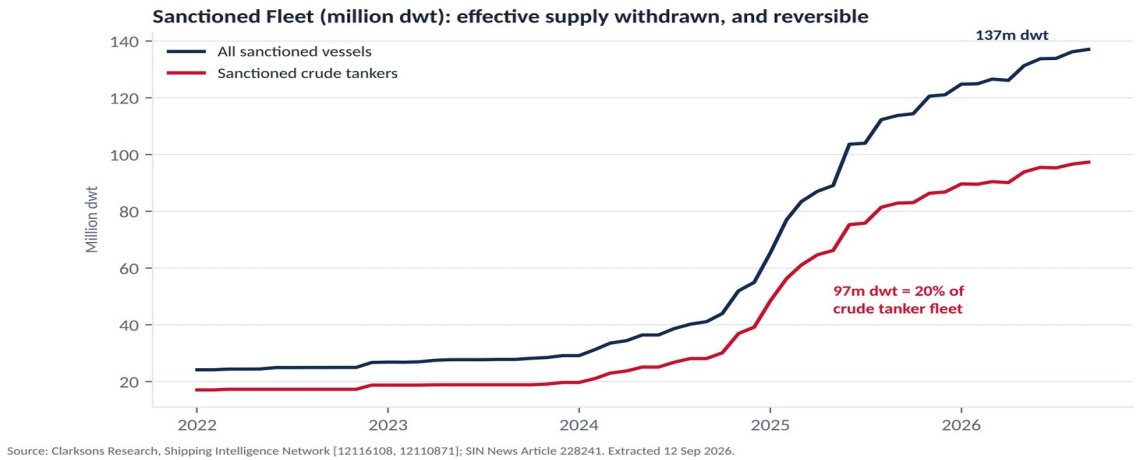
Term	Meaning
Time charter (TC)	The charterer hires the ship for a period at a daily rate and pays voyage costs; the owner pays capital and operating costs
Spot / voyage charter	One voyage at a freight rate per tonne or in Worldscale; the owner pays voyage costs
TCE	Time-charter equivalent: voyage income net of voyage costs, per day, comparable with a TC rate
Worldscale (WS)	Tanker freight index; WS100 is the published flat rate for the route, quotes are a percentage of it
Last done	The most recent transaction at the time of extraction; the starting point of charter negotiations
Orderbook	Ships contracted but not yet delivered, as a share of the fleet in dwt or in ships
Tonne-miles, average haul	Cargo tonnes times distance; average haul = tonne-miles ÷ tonnes
OPEX	Crew, insurance, stores, repair and maintenance; excludes capital and voyage costs
Eco ship	Post-2013 design with lower fuel consumption; better placed under EU ETS and FuelEU
BDI	Baltic Dry Index, a weighted average of Capesize, Panamax and Supramax time-charter averages
Kamsarmax	Bulk carrier of about 80,000 to 85,000 dwt, the largest size Port Kamsar in Guinea can load
Replacement cost	The newbuild price of an equivalent ship; secondhand above replacement cost signals over-trading

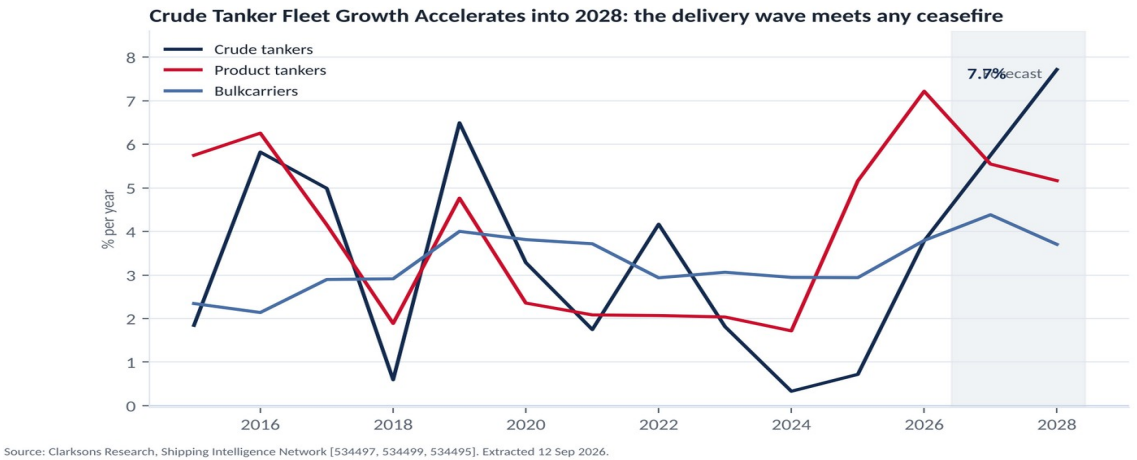
APPENDIX A10

Market backdrop: a supply shock, and a reversible one









Source: SIN series 98801 / 546177 (crude volumes and earnings), L11 (sanctioned fleet), L6 (chokepoint transits), 534438 (crude orderbook); news #221824, #227050, #228241.